Valo MCP Server → Copilot Studio Setup Guide

Connect your local Valo MCP server to Microsoft Copilot Studio using a Dev Tunnel so Copilot agents can query Snowflake semantic views.

Prerequisites

- Valo MCP server running locally (default: `http://localhost:8000/mcp`)
- Microsoft account with Copilot Studio access
- Anthropic (Claude) models enabled in your M365 tenant by an admin

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Step 1: Install Microsoft Dev Tunnel CLI

Windows:

```
winget install Microsoft.devtunnel
```

Mac:

```
brew install --cask devtunnel
```

Linux:

```
curl -sL https://aka.ms/DevTunnelCliInstall | bash
```

Step 2: Authenticate

```
devtunnel user login
```

Sign in with your Microsoft account (use the same account tied to your M365 tenant).

Step 3: Start the Tunnel

Make sure your Valo MCP server is running on port 8000, then:

```
devtunnel host -p 8000 --allow-anonymous
```

You'll see output like:

```
Connect via browser: https://abc123.usw2.devtunnels.ms:8000
```

```
https://abc123-8000.usw2.devtunnels.ms

```

Copy the second URL (the `abc123-8000` format). This routes through standard HTTPS (port 443) which Copilot Studio requires.

Your MCP server endpoint is:

```
https://abc123-8000.usw2.devtunnels.ms/mcp

```

> **Keep this terminal open.** The tunnel dies when you close it.

Step 4: Create the Main Agent in Copilot Studio

1. Go to copilotstudio.microsoft.com
2. Click **Create** → **New agent** → **Skip to configure**
3. **Name:** `Valo Analytics`
4. **Description:** `Retail analytics orchestrator that answers business questions using Snowflake semantic views via the Valo MCP server. Routes questions through a consultative workflow: clarifies the business question, constructs and executes queries, validates results, then presents findings.`
5. **Model:** Select **Claude Sonnet 4.6** (or Opus) from the model dropdown
6. **Instructions:**

```
You are a retail analytics orchestrator. Route user questions through your child agents in this order:

1. Send the question to "Valo Consultant" for clarification and analysis planning
2. Once the user confirms the plan, send it to "Valo Query Builder" for execution
3. Send the results to "Valo Validator" for QA review
4. Send validated results to "Valo Presenter" for final formatting
5. If the user requests a dashboard, send to "Valo Dashboard" for HTML generation

For simple lookups (single metric, clear scope), skip step 1 and go to Query Builder directly. Always run Validator before presenting.

```

7. **Tools:** Click **Add a tool** → connect to your MCP server using the dev tunnel URL:

```
https://abc123-8000.usw2.devtunnels.ms/mcp

```

The 5 Valo tools should auto-discover: `valo_discover`, `valo_query`, `valo_get_query_patterns`, `valo_get_memories`, `valo_query_feedback`

8. **Save** the agent

Step 5: Create the Child Agents

Create each of these as a separate agent (**Create** → **New agent** → **Skip to configure**). No tools or knowledge sources needed on child agents unless noted.

Agent 1: Valo Consultant

- **Name:** `Valo Consultant`
- **Description:** `Clarifies business questions and creates analysis plans before any data is queried.`
- **Instructions:** Paste the full content of `valo-guide/SKILL.md` and `valo-explore/SKILL.md`

Agent 2: Valo Query Builder

- **Name:** `Valo Query Builder`
- **Description:** `Constructs and executes SEMANTIC_VIEW() queries against Snowflake via the Valo MCP server.`
- **Instructions:** Paste the full content of `valo-query-semantic-views/SKILL.md`
- **Tools:** Connect this agent to the same MCP server URL so it has access to the 5 Valo tools

Agent 3: Valo Validator

- **Name:** `Valo Validator`
- **Description:** `QA reviewer for retail data analysis. Validates results and critiques reasoning before delivery.`
- **Instructions:** Paste the full content of `valo-validate/SKILL.md` and `valo-critique/SKILL.md`

Agent 4: Valo Presenter

- **Name:** `Valo Presenter`
- **Description:** `Formats validated retail analysis results for the end user.`
- **Instructions:** Paste the full content of `valo-viz/SKILL.md`

Agent 5: Valo Dashboard

- **Name:** `Valo Dashboard`

- **Description:** `Generates self-contained interactive HTML dashboards with Chart.js from validated query results.`
- **Instructions:** Paste the full content of `valo-dashboard/SKILL.md`

Step 6: Verify Child Agents Are Wired

Go back to your main **Valo Analytics** agent. Under the **Agents** section, confirm all 5 child agents appear. If they don't, add them manually.

Step 7: Test

In the Copilot Studio test panel for the main Valo Analytics agent, try these in order:

Connectivity test:

```
What data is available?

```

→ Should call `valo_discover` and return 9 semantic views.

Consultative test:

```
How are pants doing?

```

→ Should ask clarifying questions, NOT query immediately.

Full chain test:

```
Pants is the CLASS value. All channels, February 2026. Net sales by fiscal week and channel. Go ahead.

```

→ Should query, validate (check for truncation, flag anomalies), and present with confidence rating.

Dashboard test:

```
Now build me a dashboard for that.

```

→ Should generate self-contained HTML with Chart.js charts, KPI cards, and embedded data.

Troubleshooting

Problem	Fix

Copilot can't connect to MCP server	Check the dev tunnel terminal is still running. Restart with `devtunnel host -p 8000 --allow-anonymous`
Only GPT models in dropdown	Admin needs to enable Anthropic models: M365 Admin Center → search for Anthropic → opt in
Tools not discovered	Verify URL ends with `/mcp`. Try `/sse` as an alternative.
Query Builder skips `valo_get_memories`	Strengthen instructions: add "MANDATORY FIRST STEP: call valo_get_memories before every query" at the top
Dashboard HTML blank in browser	Copy the code block from chat carefully — ensure you get the complete content from `<!DOCTYPE html>` through `</html>`

Notes

- The dev tunnel URL changes each time you restart. Update the MCP server URL in Copilot Studio if you restart the tunnel.
- For a persistent URL, use `devtunnel host -p 8000 --allow-anonymous --tunnel-id <your-id>` after creating a named tunnel with `devtunnel create`.
- The Valo MCP server requires no modifications — it's client-agnostic. The same server works with [Claude.ai](https://claude.ai), Copilot Studio, or any MCP-compatible client.